

09 — Source Code, API Spec & Installation

Source code

All source code is in this repository (the working app lives at the repo root, not inside `submission/`):

- **Frontend:** `../frontend` — vanilla HTML/CSS/JS, one file per screen, plus `api.js` (API client) and `app.css`.
- **Backend (microservices):** `../backend/lambda`s — 13 AWS Lambda handlers (one file per operation) plus a Cognito Post-Confirmation trigger (`auth_post_confirmation`) and a shared `_common.py` auth helper. Every handler is **self-documented** with a `Feature: Fxx` docstring and inline validation comments.
- **Provisioning / deploy scripts:** `../backend/deploy` and `../frontend/deploy_frontend.py`.
- **One-command orchestrator:** `../redploy_all.py`.
- **Database scripts:** `../database`.

API specification (Swagger / OpenAPI)

The full API is documented as an **OpenAPI 3.0** spec covering all 13 routes (paths, methods, parameters, request/response schemas, status codes, and the admin security scheme):

- **Spec file:** `../backend/api/openapi.yaml`
- **Browsable Swagger UI:** `../backend/api/index.html` — open it through a local web server so the browser can load the spec:

```
cd backend/api
python -m http.server 8000
# then open http://localhost:8000/
```

- You can also paste `openapi.yaml` into <https://editor.swagger.io> to view it.

A human-readable companion reference is in `../docs/api.md`.

Installation instructions

Full instructions are in `../docs/installation.md`. The short version (from the **repository root**, region `us-east-1`):

```
# 1. install the single dependency (CloudShell already has it)
pip install -r requirements.txt

# 2. deploy the whole serverless stack and load demo data
python3 redeploy_all.py --seed
```

The script prints the **Website URL**, the **API URL**, and a **demo host link**. Other modes:

Command	What it does
<code>python3 redeploy_all.py</code>	Deploy without demo data
<code>python3 redeploy_all.py --seed</code>	Deploy and load demo data
<code>python3 redeploy_all.py --clean --yes --seed</code>	Wipe, then deploy fresh with demo data
<code>python3 redeploy_all.py --teardown --yes</code>	Remove all SeatMe AWS resources

What gets deployed

1. DynamoDB table `SeatMe`.
2. 13 API Lambda functions + the Cognito Post-Confirmation trigger + API Gateway HTTP API (`setup_aws.py`).
3. Cognito user pool + app client + `host/admin` groups + Post-Confirmation trigger wiring + seeded admin/host users (`setup_cognito.py`).
4. (with `--seed`) demo data (`seed_example.py`).
5. Frontend uploaded to S3 with the API URL and Cognito IDs injected (`deploy_frontend.py`).

Self-documentation

- Each Lambda begins with a docstring naming its **Feature** and route.
- `frontend/api.js` documents every route→Lambda mapping at the top of the file.
- Validation rules are commented inline in each handler.